Yanshan Jin

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EDUCATION:

University of California, Davis **Davis, CA Sep 2023 – Present**

Ph.D. Candidate in Nutritional Biology, Research focus: health informatics and precision nutrition
Advisor: Angela M. Zivkovic

Zhejiang University **Zhejiang, CHINA Sep 2020 – Jun 2023**

M.Sc. in Animal Science, State Key Laboratory of Molecular Animal Nutrition

Advisor: Hongyun Liu

Nanjing Agricultural University **Jiangsu, CHINA Sep 2016 – Jun 2020**

B.Sc. in Animal Science

RESEARCH INTERESTS:

Precision nutrition, diet–microbiome interactions, and AI-assisted dietary assessment; metagenomic analysis and computational approaches to nutrition and aging.

CONFERENCE:

- **Jin Y**, …, Zivkovic AM. *Machine Learning-Based Prediction Models for Alzheimer's Disease Diagnosis Using Serum Glycopeptides.* **Bay Area Alzheimer's Disease Researchers Symposium.** June 7, 2024; Davis, CA. Poster.
- **Jin Y**, Zivkovic AM, …, Yiu G. *Impact of Diet and Gut Microbiome on Nonhuman Primate Models of Age-Related Macular Degeneration.* **ASN NUTRITION 2025**. Poster presentation; **Current Developments in Nutrition.**
- Sun Q, **Jin Y**, …, Zivkovic AM. *NMR-Derived Lipoprotein Remodeling and Macular Pigment Response in Older Adults at Risk for AMD: Interim Analysis of the Goji Berry Trial.* **ASN NUTRITION 2026**. **Current Developments in Nutrition.**

PUBLICATIONS:

- **Jin Y**, Zivkovic AM, …, Yiu G. *Impact of Diet and Gut Microbiome on Nonhuman Primate Models of Age-Related Macular Degeneration.* **Translational Vision Science & Technology** (under review)
- Jing N, …, **Jin Y**, …, Zivkovic AM, …, Zhang G. *Gut Microbiota Mediate the Metabolism of Colonic Prostaglandins.* **Nature**. (under review)
- Romo EZ, …, **Jin Y**, Kang JW, Zivkovic AM. *A low-dose prebiotic fiber supplement reduces lipopolysaccharide-binding protein concentrations in a subgroup of young, healthy adults consuming low-fiber diets.* **Nutrition Research.** 2025;133:138–147.
- **JIN Y**, …, Zivkovic AM. *Sequencing-depth sensitivity of diet-relevant microbial functions across annotation and profiling strategies.* Manuscript in preparation.

RESEARCH EXPERIENCE:

AI-Assisted Dietary Assessment **Nov 2025 – Present**

- Develop an AI-assisted food database integration workflow for microbiome-focused dietary assessment.
- Design pipeline evaluation protocols and lead a registered dietitian team in performance assessment.
- Developed an AI-assisted food diary digitalization tool and supervised undergraduates for tool evaluation.
- Map dietary records to microbial substrates, including oligosaccharides, polyphenols, and other bioactives.

Sequencing Depth Benchmarking for Metagenomic and Metatranscriptomic Profiling Nov 2025 – Present

- Benchmark sequencing depth needed to detect microbial functions targeting specific dietary substrates.
- Compare analytical strategies to support cost-efficient analysis of diet-microbiome interactions.
- Evaluate the preservation of individual longitudinal microbial responses at lowered sequencing depth for precise nutritional intervention.

Goji Berry Supplementation and Retinal Health Clinical Trial Sep 2023 – Present

- Contribute to trial design and planning with ophthalmologists, hospital study coordinators, and the university's Clinical Trial Ccenter.
- Analyzed Electronic Medical Record data to identify older adults at elevated risk of age-related macular degeneration (AMD) for participant recruitment.
- Manage study visits and the collection of dietary and clinical data, blood, and stool samples.
- Build REDCap data capture instruments and a customized online participant management system.
- Contribute to the designing and planning of an additional pilot clinical trial to optimize goji berry bioactive delivery approaches.

Diet and Gut Microbiome in a Primate Model of AMD Sep 2024 – Sep 2025

- Performed bioinformatics and statistical analyses of metagenomic data to compare microbial composition and functional genes across diets and retinal pathology groups.
- Interpreted and visualized findings and contributed to manuscript writing and revision; presented a poster at ASN NUTRITION 2025.

SKILLS:

- **Programming and Scientific Computing:** R, Python, Linux, high-performance computing, Snakemake, GitHub, ShinyApp, Machine Learning.
- **Bioinformatics and Omics:** Metagenomics, metatranscriptomics, transcriptomics, metabolomics, glycomics, and NMR lipidomics.
- **Dietary Assessment:** Food database crosswalks, dietary record digitization, commercial food decomposition
- **Research Methods:** Study visit coordination; dietary and clinical data collection; scientific visualization; manuscript and proposal writing; undergraduate training.

FELLOWSHIPS AND AWARDS:

- Kosuna Doctoral Fellowship, UC Davis, Nov 2026
- Jastro and Shields Graduate Research Award, UC Davis, Jul 2026, Jul 2025, Jul 2024
- GGNB Professional Development Fellowship, UC Davis, May 2026
- Schneeman Research Award, UC Davis, Feb 2026
- Campbell Research and Travel Award, UC Davis, Jan 2025
- Best Oral Presentation Award, International Annual Dairy Animal Lactation Seminar, Apr 2023
- M.Sc. Academic Scholarship (3-year Full tuition & stipend), Zhejiang University, 2020–2023
- Excellent Undergraduate Award, Nanjing Agricultural University

TEACHING AND RESEARCH LEADERSHIP:

- Served as a teaching assistant for NUT 11 Current Topics & Controversies in Nutrition and NUT 116B Clinical Nutrition across multiple quarters
- Led discussion sections, held office hours, and graded assignments with individual feedback.
- Recruited, trained, and supervised 10 undergraduate interns in clinical research methods.
- Managed undergraduate team and collaborations, assigned research tasks, coordinated evaluation activities, and reviewed undergraduate outputs for accuracy and consistency.